Comments to the author:

Reviewer: 1

Comments to the Author

1. Please clarify the privacy and threat model because federated learning alone does not guarantee privacy.
2. The results rely on analytically generated data so please validate performance under measurement noise and EMT scan uncertainty.
3. Please quantify client heterogeneity and test FedAvgIm under more severe non IID distributions.
4. Please report communication overhead including model size number of rounds and total uplink and downlink cost to support practicality claims.
5. Please extend the stability validation to multiple operating points and report the stability decision consistency.
6. The paper uses impedance identification while the model outputs admittance so please unify the terminology for consistency.

Reviewer: 2

Comments to the Author

The manuscript proposes a FedAvgIm to address data scarcity and privacy protection in the impedance identification of IBRs. While the application of decentralized learning to this domain is noted, the study relies on a fundamental methodological contradiction: utilizing analytical models to generate training data for a problem specifically motivated by the need for "black-box" identification. This logical disconnect, combined with the restrictive assumption that all participating IBRs share an identical small-signal structure, significantly limits the practical engineering value and generalization of the proposed framework. These core issues, along with concerns regarding model architecture and error sensitivity, must be addressed. The specific comments are as follows.

1. The primary motivation stated in the Introduction is to identify impedance for "black-box" inverters where internal parameters are unknown. However, in Section III-A, the authors state that the training datasets are generated using analytical models based on the component-connection method. If the analytical models are already available to generate the training data, there is no engineering need to train a Neural Network to "identify" the impedance. The authors should either justify this approach or, more preferably, validate the framework using measurement-based data to faithfully represent the "black-box" scenario

described in the motivation.
2. The manuscript assumes that the nine participating IBRs "share the same small-signal structure but differ in their parameters". In practical multi-site smart grids, IBRs are highly heterogeneous, involving different vendors, topologies and control strategies. The standard FedAvg algorithm often struggles with non-IID data arising from such structural heterogeneity. The authors need to discuss the applicability of FedAvgIm in a realistic environment with mixed IBR structures. A discussion on this limitation is necessary to align the contribution with practical grid conditions.
3. In Section III-B, Equation (5) describes a "Trunk + Head" architecture inspired by personalized federated learning. However, the text immediately states that "both the trunk and head parameters are trained globally". This appears contradictory. If the entire network is aggregated globally, the distinction between trunk and head is redundant. The authors should clarify the purpose of this architectural split. If personalization was intended to address the heterogeneity mentioned above, it should be implemented and compared against the fully global approach.
4. In Case 4, the estimated admittance is used for Nyquist stability analysis. Since neural networks inevitably produce prediction errors, the authors should provide a sensitivity analysis or error bounds. Specifically, how sensitive is the stability criterion to small prediction errors in the phase or magnitude of Ydq near the crossover frequency? A discussion on uncertainty quantification would significantly strengthen the reliability of the proposed stability assessment.
5. The paper mentions using a full frequency set for testing but subsampling for training. Given that impedance responses typically exhibit wide variations over logarithmic frequency scales, the distribution of the sampled frequencies in the training set is critical. The authors should clarify the sampling strategy for the frequency input f. Using linearly spaced samples for training might bias the model towards high frequencies and neglect critical low-frequency dynamics.

Reviewer: 3

Comments to the Author
Thank the authors for their hard work. Overall, I find this work valuable. My comments and questions are as follows:
1. A sudden step change of grid impedance with 50% increase and the following instability seems unrealistic. Please find a way to justify it or use a more realistic scenario.
2. Please clarify the benefits of the proposed approach compared to the parametric impedance model.
3. Please state if there is any methodological novelty beyond FedAvg.

4. What are the limitations to scale up the method with many IBRs?
5. Please include a comparison of communication complexity and cost, as well as a comparison of the required computational power.
6. Is the dataset realistic and noisy, or clean simulation results? Please state that in the manuscript.

Reviewer: 4

Comments to the Author
The manuscript provides a novel and well-structured approach to federated learning in impedance identification, with convincing theoretical development and experimental validation.

1. In Case 3, the local transfer learning model is pretrained only on IBR4 (100 samples), whereas the federated transfer learning model leverages data from IBR2–IBR9 (1670 samples in total). Since the pretraining data scale differs substantially between the two settings, it is unclear whether the reported performance improvement stems from the federated mechanism itself or from the larger and more diverse pretraining dataset. It would strengthen the manuscript if the authors could clarify this point or provide an additional controlled experiment where the total pretraining data size is comparable across methods.

2. In Case 4, the stability analysis is conducted under a single operating condition. Could the author clarify whether prediction errors might lead to a potential misinterpretation of stability?

3. The manuscript notes that the framework is potentially applicable to grid-forming inverters, however, could the proposed FedAvgIm framework be extended to mixed GFL–GFM systems?

Reviewer: 5

Comments to the Author
Are the dq admittance data estimated using the proposed method accurate enough for direct application in small-signal stability analysis? The article provides only one example. If numerical errors in the admittance lead to incorrect estimation results in practical

applications, what would be the consequences? Can the confidence level of the stability prediction results be quantified?

Additionally, at the end of the first chapter, a logical overview of the remaining chapters should be provided in accordance with standard writing conventions.